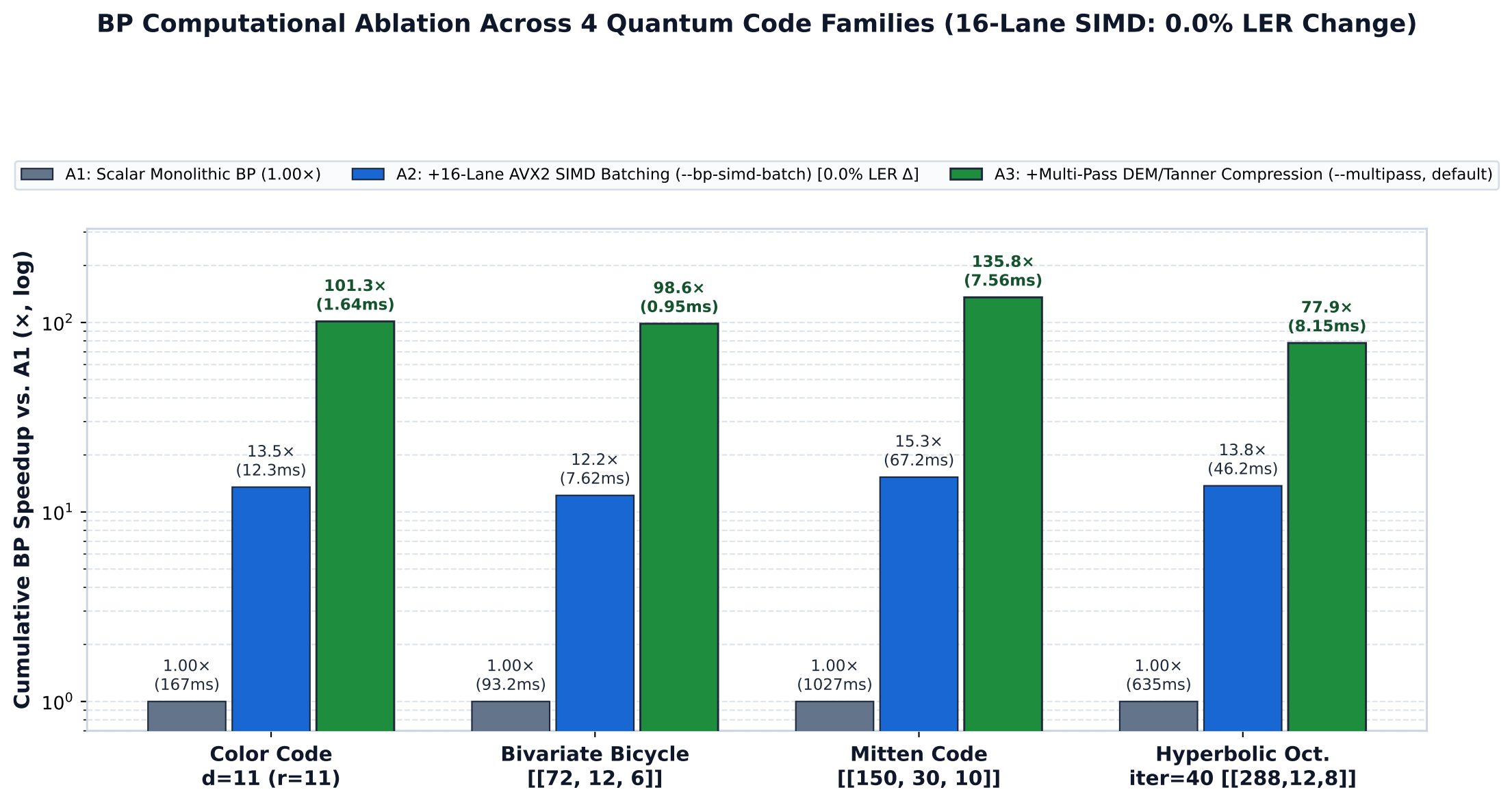
BP Computational Ablation Across 4 Quantum Code Families (16-Lane SIMD: 0.0% LER Change)
A1: Scalar Monolithic BP (1.00×)
A2: +16-Lane AVX2 SIMD Batching (--bp-simd-batch) [0.0% LER Δ]
A3: +Multi-Pass DEM/Tanner Compression (--multipass, default)
Cumulative BP Speedup vs. A1 (×, log)
10^2
10^1
10^0
1.00× (167ms)
13.5× (12.3ms)
101.3× (1.64ms)
1.00× (93.2ms)
12.2× (7.62ms)
98.6× (0.95ms)
1.00× (1027ms)
15.3× (67.2ms)
135.8× (7.56ms)
1.00× (635ms)
13.8× (46.2ms)
77.9× (8.15ms)
Color Code d=11 (r=11)
Bivariate Bicycle [[72, 12, 6]]
Mitten Code [[150, 30, 10]]
Hyperbolic Oct. iter=40 [[288,12,8]]